CO1 - AT1

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Course Code : CSA 5111

Course Name : Cryptography and Network Security

Scenario: Event communication Security

A Student Organisation is preparing to share onfidential event detain with its core members Using a simple classical encryption method. During testing. a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an all thorized member aware of the encoding rule. would remove the original message.

Caesar cipher.

The caesar cipher is classical Substitution encryption technique in which each letter in the plaintext is Shifted by a fixed number of position in alphabet. In this Scenario, the Mesage "MEET AT NOON" is encrypt using a Shift of 4 position.

a) Encrypt the message using a Shift of 4 Positions

Plaintext : MEE ' A T NOON

Shift key : K = 4

Encryption Method : caesar cipher

$$C = (P + K) \text{ Mod } 26$$

C - Cipher text letter value

P - Plaintext letter value

K - Shift key

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25

Limit

(12 4 4 19 / M E E T) (0 19 / A T) (12 14 14 13 / N O O N)

(Q I I X) (E X) (R S S R)

$C(M) = (12+4) \% 26$

$= 16 \% 26 = 16 = Q$

$C(E) = (4+4) \% 26$

$= 8 \% 26 = 8 = S$

$C(T) = (19+9) \% 26$

$= 23 \% 26 = 23 = X$

$(A) = (0+4) \% 26$

$= 4 \% 26 = 4 = E$

$C(N) = (13+4) \% 26$

$= 17 \% 26 = 17 = R$

$C(O) = (14+4) \% 26$

$= 18 \% 26 = 18 = S$

Cipher text = QIIX EX RSSR

b) How an authossed member decrypts the message

An authoxred member who knows the Shift Key(A) can decrypt the cipher text by shifting each letter four

Limitations:

1) It has only 25 possible non-zero Shift key: making brute - force attacks easy.

2. Letters frequency pattern remain Unchanged

3. It is not suitable for protecting highly Confidential information in Modern Communication System

Scenario: Intercepted Suspicious Message.

A CyberSecurity intern the coded Message: "GSRH RHZ HVXIVG" during network monitoring. The Structure of the text Suggests a Simple Substitution mechanism rather than Complex encryption.

a. Deduce the most likely Original Message

b) Justify your answer by analyzing repeating patterns. letter Structures or Known Cipher Characteristics.

a) Decryption

The intercepted ciphertext is "GSRH RHZ HVXIVG".

This message can be decrypted using Monoalphabetic Substitution cipher the reverse-alphabet Substitution

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L

P	Q	R	S	T	U	V	W	X	Y	Z
K	J	I	H	G	F	E	D	C	B	A

Positions backward in the alphabet

Decryption formula:

$$P = (C - K) \% 26$$

16 8 8 23 4 23 17 13 13 5
(QIIX EX RSSF)
MEET AT NOON

$P(Q) = (16 - 4) \% 26$

$= 12 \% 26 = 12 = M$

$P(I) = (8 - 4) \% 26$

$= 4 \% 26 = 4 = E$

$P(X) = (23 - 4) \% 26$

$= 19 \% 26 = 19 = T$

$P(E) = (4 - 4) \% 26$

$= 0 \% 26 = 0 = A$

$P(R) = (17 - 4) \% 26$

$= 13 \% 26 = 13 = N$

$P(S) = (18 - 4) \% 26$

$= 14 \% 26 = 14 = O$

Plain text : MEET AT NOON

Advantages.

1. Simple to understand and implement.
2. Require only fixed Shift key.
3. Useful for learning basic encryption and decryptions concepts.

GSRH RH Z HVXIVG
THIS IS A SECRET

Plain text : This is A SECRET

Justification

① Simple substitution Mechanism: The ciphertext contains uppercase letters and Spaces, Suggesting that word boundaries have been preserved. this is consistent with a classical substitution cipher.

② letter-by-Letter Replacement: Every letter is replaced day its reverse alphabet counterpart. for G → T, S → H R → I and H → S, Therefore, GSRH → THIS

③ Word Structure: The ciphertext contains 4 words with length 4, 2, 1, 6. The decrypted Message, This is A SECRET, has exactly the same - word length patten.

④ No Secret key required: It uses a fixed alphabet reverse rule, so Sender. and recived do not need to exachange a Separations encryption key.

) Scenario: Secure Messaging System:
A company deploys a keyword. based encryption technique for internal communication. An employee sends the word "MELLO" using the keyword "KEY" to ensure confidentiality
a) Denion Stocte how the encryption proces transforms the Message Step-by-Step.

b) Explain how the repeating keyword influences each cha of ciphertext.

a) Encryption process step-by-Step

The Vigenere cipher is a polyalphabetic Substitu cipher that uses a repeating keyword to encrypt a plain text message.

Given:

*) plaintext : HELLO *) Keyword : KEY

*) Encryption Method : Vigenere cipher.

$$C = (P+K) \% 26$$

$$C(H) = (H+K) \% 26$$
$$= (7+10) \% 26$$
$$= 17 \% 26 = 17 = R$$

$$C(E) = (4+4) \% 26$$
$$= 8 \% 26 = 8 = I$$

$$C(L) = (11+24) \% 26$$
$$= 35 \% 26 = 9 = J$$

$$C(L) = (11+10) \% 26$$
$$= 21 \% 26 = 21 = V$$

$$C(O) = (14+4) \% 26$$
$$= 18 \% 26 = 18$$
$$= S$$

Ciphertext : RIJVS

7	4	11	11	14
H	E	L	L	O
K	E	Y	K	E
10	4	24	10	4

H E L L O
R I J V S

Each Chapter

How the repeating keyword influences each character.

1) Keyword Repetition: The keyword KEY is repeated as KEYKE to match the length of the plain text.

2) Different Shifts: Each keyword letters determine the Shift applied to its corresponding plaintext letters. for example K Shifts H by 10 position, while E Shift E by 4 position.

3) Multiple Substitution alphabets: Unlike two square cipher which uses 1 fixed shift, Vigenere cipher uses different Shift based on the keyword letters.

4) Improved Security: The repeating keywords makes the encryption pattern less obvious than a simple caesar cipher. although Short or repeated keywords can still be Vunerable to cryptanalysis.

5) Decryption: The receiver uses the same keyword and Subtracts the key value from each ciphertext letters using:

$$P = (C - K) \bmod 26.$$

This recover the original Message HELLO.

4) Scenario: Surpicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a) Analyze how investigators could detect the presence of hiidden data within the image.

b) Recomnaed two techniques that could Strengthen the Security of Such hidden communications against detection or extraction.

a) Methods to detect hidden data in an image.

IMAGE STEGNOGRAPHY is a technique used to wide confidential information unside an image file without visibly changing its appearance. Investigators can use the following Methods to detect hidden data.

1. Visual Inspection: Compare the original image with Suspected Image and Examine unusual colors, pixels or Patter. Hidden datas may not be visible to naked eye.

2. File Size analysis: check whether the image file size is unusally large compared with similar image of same resolution and format. Extra hidden information may increase the file size.

3. Meta data analysis: Examine metadata such as file format dimensions, time stamp and comments using forensic tools. Unusual metadata may indicate Suspicious activity

4. Statistical Analysis: Analyze pixel values, color distribution and bit patterns. Stegacography may introduce statical irregularities that Suggest hidden information.

Steganography tools : Use forensic tools such as Stag Expose or Stegdetect, when applicable, to identify suspicious patterns associated with certain Steganographic Methods.

b) 2 technique to Strengthen the security of hidden communication.

1. Encryption before embedding.

Encrypt the confidential message using a strong encryption algorithm such as AES before embedding it into the image.

→ Even if investigators extract the hidden data, they cannot easily understand it without the decryption key

→ It provides an additional layer of confidentiality

→ It protects the message even if the steganography method is discovered.

2. Least Significant bit (LSB) Steganography with randomized embedding.

Embed encrypted data in selected pixel bits using a secret pseudorandom embedding pattern rather than storing it sequentially

→ It can make hidden data harder to locate through simple visual inspection

→ Randomized positions can reduce obvious patterns associated with sequential empedding.

→ The receiver needs the correct Secret key or pattern to locate and extract the message.

f) Scenario: Legacy Encryption in Corporate communication,
A legacy system in a company still uses a transposition
-based encryption method with three levels of arrangement
A received message appears as:

"WECRLTEERDSOEFFEAOCAIVDEN".

a) Determine the most probable original message by reversing the encryption process.

b) Critically evaluate why such a method is vulnerable in Modern communication Systems.

c) propose one practical enhancement to improve the confiden tiality of message in this Scenario.

Decryption:

Given Ciphertext: "WECRLTEERDSOEEFEAOCAIVDEN"

The Rail Fence Cipher is a transposition cipher that rearranges letters in a zig zag pattern across three rails. The exact plain text cannot be confirmed without verifying the encryption pattern and rail arrangement.

Decrypt plaintext: WE ARE DISCOVERED FLEE AT ONCE using 3-Rail Fence Cipher.

Vulnerabilities:

1. Small key Space: Easy to break by trying different rail arrangements.
2. Letter Frequency: Original letter Frequencies.
3. Weak Security: Does not provide sufficient confidentiality against modern attack.

, one practical enhancement:

use AES encryption before transmission.

The company should replace the legacy transpositiononly method with AES-256, a modern symmetric encryption algorithm. The sender encrypts the original message using a secret key and the authorized receiver decrypts it using the corresponding key.

advantages:

→ Provides strong confidenticality

→ Makes unacthorized decryption computationally impractical when implemented correctly.

→ can be used with anthenticated encryption mode such as AES-GCM to detect tampering.